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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

November 2018 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-NOV18

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

November 2018

Insert

Instructions
- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice
- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the Question Paper/Answer Book.

4368

QAN 500/8702/2

IB/M/Nov18/E4

Data Sheet for Angling

Anglers

Anglers try to catch fish.



In competitions, anglers try to catch the biggest total weight of fish.

Angling competition

Some competitions start with quarter-finals, then have semi-finals and a final.

Each quarter-final has 9, 12 or 15 anglers.

The table shows the number of anglers who qualify for a semi-final.

Quarter-finals	
Number of anglers in quarter-final	**Number of anglers who qualify for semi-final**
9	3
12	4
15	5

Qualifiers from each semi-final take part in the final.

Each semi-final has 12, 16 or 24 anglers.

The table shows the number of anglers who qualify for the final.

Semi-finals	
Number of anglers in semi-final	**Number of anglers who qualify for final**
12	3
16	4
24	6

Weight

The weight of each fish can be measured in

pounds (lb) and ounces (oz)

or

kilograms (kg) and grams (g).

1 lb = 16 oz

Converting ounces to grams

To convert a weight from ounces to grams, use the formula

$y = 28.35x$

where y is the weight in grams

x is the weight in ounces

Example

Convert 376 ounces to grams.

$28.35 \times 376 = 10\,659.6$

376 ounces is 10 659.6 grams.

Data Sheet for Council Tax

If you own or rent your home, you usually have to pay Council Tax.

There are eight bands of Council Tax, from **A** (lowest) to **H** (highest).

The more valuable your home, the higher your band and the more money you have to pay.

The table shows the council tax **per home** for each band in three areas of Waybridge.

	Council Tax bands (£ per year per home)							
	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
Heath	1121.59	1308.51	1495.44	1682.38	2056.23	2430.10	2803.96	3365.76
Mossett	986.59	1151.01	1315.45	1479.89	1808.74	2137.61	2466.47	2959.77
Kribly	1190.85	1389.31	1587.79	1786.27	2183.21	2580.16	2977.11	3572.54

Council Tax covers the period from April 1st of one year to March 31st of the next year.

In Waybridge, you can pay your Council Tax in 10 monthly payments from April to January.

You normally pay the full Council Tax if you are aged 18 or over.

You get a 25% **discount** if you are the **only** person aged 18 or over living in your home.

Example

Tina is 32 and lives on her own in Kribly.

The Council Tax band for her home is band C.

The band C amount for Kribly is £1587.79 per year.

Tina gets a 25% discount, so she pays £1190.84 per year.

She pays in 10 monthly payments.

$1190.84 \div 10 = 119.084$

This is rounded to the nearest pound.

An adjustment is made in April to give the correct total for the year.

$1190.84 - 119 \times 9 = 119.84$

Tina pays £119.84 in April.

She then pays £119 each month from May to January.

Disabled people

If you are disabled and your home is adapted for your needs, your home is re-banded.

It is re-banded to one band **lower than** the usual band for that home.

You can still get the discount for living alone.

Use of Council Tax in Waybridge

Looking after children	15%
Education	8%
Road repairs and street lighting	7%
Buses	8%
Care for vulnerable adults	32%
Homelessness	5%
Arts and leisure	7%
Bin collection	4%
Street cleaning and flood defences	2%
Trading standards and food safety	2%
Planning	3%
Other services	7%

END OF DATA

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Answer **all** questions in the spaces provided.

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1 Angling

There is a **data sheet** for Angling.

1 (a) In an angling competition there are seven quarter-finals.

The table shows the number of anglers in each of the quarter-finals.

Q1 Worked Solution:

See official AQA mark scheme method.

Quarter-final 1	15 anglers	Semi-final 1	Final
Quarter-final 2	12 anglers		
Quarter-final 3	12 anglers		
Quarter-final 4	9 anglers		
Quarter-final 5	12 anglers	Semi-final 2	
Quarter-final 6	12 anglers		
Quarter-final 7	12 anglers		

How many anglers qualify for the final?

You **must** show your working.

[4 marks]

Question 1 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

[Method & Accuracy]

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1 (b) John and Lata are in one of the quarter-finals.
Here are the weights of the fish they have caught in the first 12 hours.

John	17 lb 2 oz	19 lb 12 oz	16 lb 13 oz	20 lb 5 oz
Lata	18 lb 4 oz	17 lb 1 oz	18 lb 11 oz	

John says,

"The mean weight of my fish is **more than** the mean weight of Lata's fish."

Is he correct?
You **must** show your working.

[5 marks]

Do not write outside the box

1 (c) The weight of the heaviest fish that John catches is 20 lb 3 oz
Last year, in Australia, he caught a fish weighing 9 kg 120 g
John says,

"The fish I caught in Australia was heavier."

Is he correct?
You **must** show your working.

[5 marks]

14

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2 **Council Tax**

There is a **data sheet** for Council Tax.

2 (a) Tom and Mary are both 47 years old.

They share a home in Kribly.

The Council Tax is band **G**.

They pay their Council Tax in 10 monthly payments.

Work out the Council Tax payment for April.

[3 marks]

Q2 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Question 2 continues on the next page

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2 (b) Dan is disabled and lives on his own.
His home is **not** adapted for his needs.
He moves to a new home.

He makes these notes.

	Old home	New home
Area	Heath	Mossett
Usual band	**C**	**D**
25% discount	Yes	Yes

Dan will adapt his new home in Mossett for his needs.

He says,

"My Council Tax will then be cheaper than I paid in Heath by at least £100"

Is he correct?
You **must** show your working.

[6 marks]

2 (c) In Waybridge, the total income from Council Tax is £38 320 000

Estimate how much of the £38 320 000 the council spends on buses each week.
Give your answer to a suitable degree of accuracy.

[3 marks]

Do not write outside the box

David is calculating the Council Tax in Bretton, another part of Waybridge.

The Council Tax for band **D** is £1504.44

He works out the Council Tax in the other bands using these proportions.

	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
Proportion of band **D**	$\frac{6}{9}$	$\frac{7}{9}$	$\frac{8}{9}$	1	$\frac{11}{9}$	$\frac{13}{9}$	$\frac{15}{9}$	$\frac{18}{9}$

†2 (d) Work out the Council Tax in band **B**.

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

2 (e) How many times bigger is the Council Tax in band **H** than in band **A**?

Circle your answer.

[1 mark]

$\frac{12}{9}$ 2 3 12

16

Turn over for the next question

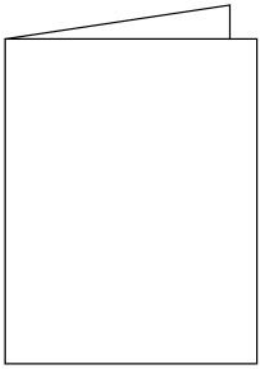
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3 Christmas Cards



I make and sell Christmas cards.
I put designs onto card blanks.

Leah



Card blank



Card blank with design

Do not write outside the box

†3 (a) You buy card blanks in packs.

Card Blanks
Packs of 15
Only £6.49

Work out the cost of 60 of these card blanks.

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 3 continues on the next page

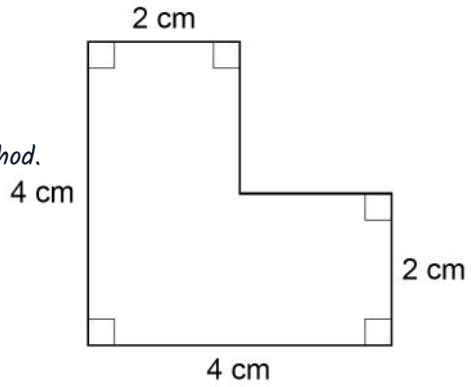
Do not write outside the box

Leah is designing a card with a chimney and a **pair** of Santa's boots.



3 (b) Leah cuts the boots from sheets of paper measuring 13 cm by 10 cm

Here is a sketch of one of the boots.



Not drawn accurately

Q3 Worked Solution:

See official AQA mark scheme method.

[B1] [Aa]

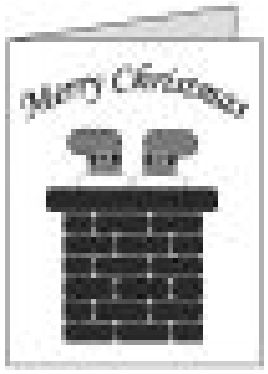
Show how 10 boots can be cut from a 13 cm by 10 cm sheet.

Use the grids opposite.

[3 marks]

Do not write outside the box

3 (c) Leah makes 150 cards with this design.



She uses 15 sheets of red felt for the chimneys.

She buys a bulk pack of 150 card blanks.

This table shows the costs of the items she needs for the cards.

Item	**Cost**
Paper for boots	75p per sheet
Red felt	£1.40 per sheet
150 card blanks	£43.50
Other costs	£16.50

Leah sells each card for £1.20

She says,

"My profit will be **more than** 65% of the cost of making the cards."

Is she correct?

You **must** show your working.

[8 marks]

Do not write outside the box

Jane and her family are planning a visit to Winter Wonderland.

4 (a) Here are the ticket prices.

Ticket Prices

£38 per adult or child aged 5 and over

£25 per child aged under 5

plus £5 booking fee per family

Q4 Worked Solution:

In particular, individual marks are mapped onto the following skills standards.

Altogether, how much will Jane pay for herself and her family?

[3 marks]

Representing
Making sense of the situations and representing them.
A learner can:
Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.
Identify the situation or problems and identify the mathematical methods needed to solve them.
Choose from a range of mathematics to find solutions.
Analysing
[Ra] [Rb] [Rc] [Aa] [Ab]

4 (b) The activities at Winter Wonderland start at 10 am

Jane and her family want to arrive 45 minutes before the first activity starts.

They live 80 miles away.

Jane will drive at an average speed of 50 miles per hour.

What is the **latest** time they should set off?

[4 marks]

Do not write outside the box

4 (c) Jane has already booked the following activities.

Time	Activity	Length of activity
10 00	Christmas theatre	40 minutes
13 30	Mrs Christmas	40 minutes
16 10	Elves	30 minutes

She has also booked lunch at the Boat Inn from 12 30 to 13 20

Jane is planning other activities for the rest of the day.



We need at least 10 minutes between activities and before and after lunch.

We want to do as many different activities as possible.

We will leave after the Elves activity.

Here are the other activities.

Each activity can start at any time.

Activity	Length
Snow tubing	40 minutes
Reindeer	30 minutes
Ice skating	50 minutes
Sleigh ride	20 minutes
Winter crafts	30 minutes
Treasure hunt	40 minutes

Do not write outside the box

4 (d) Myra and some friends are going to Winter Wonderland.

They will stay for 3 nights.

Myra looks at the prices of different-sized lodges.

Arrival date	**Size of Lodge**		
	Sleeps 4	Sleeps 5	Sleeps 6
Friday 23rd Nov (3 nights)	£320	£385	£455
Friday 30th Nov (3 nights)	£378	£439	£495
Friday 7th Dec (3 nights)	£410	£450	£545

Myra is booking lodges for 20 people.

They will arrive on Friday 30th November.

What is the cheapest possible price for her booking?
State clearly how many of each type of lodge to book.

[4 marks]

16

END OF QUESTIONS

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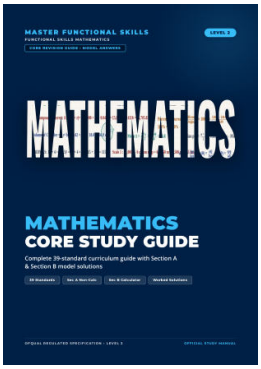
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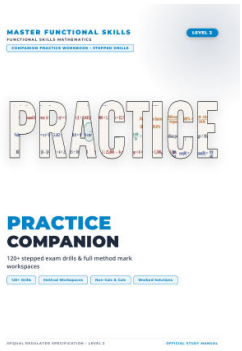


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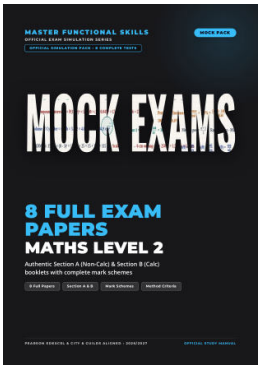


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